

Discounted Dynamic Reputation Investment with Hidden Product Choice

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We study a discounted reputation problem in which a seller privately chooses between a low-cost product and a high-quality product while a user learns only from realized binary outcomes. The user treats the seller as a single Bernoulli arm and applies Thompson sampling to decide whether to engage with the seller or with a known outside option. The seller chooses hidden products to maximize expected discounted profit. We formulate the infinite-horizon Bellman equation and solve a long-horizon temporal approximation by exact backward induction over all reachable posterior states. The implementation imposes no artificial boundary on the number of observations. The optimal product choice is governed by the discounted continuation value of a success relative to a failure. We also derive a deterministic discounted fluid approximation that preserves this threshold structure and identifies a long-run quality-mix benchmark. Numerical experiments with a patient seller illustrate how reputation investment varies with the quality of the outside option.

Key words: dynamic reputation; hidden actions; discounted dynamic programming; fluid approximation; Thompson sampling

1. Model Formulation

Time is discrete and indexed by $t = 1, 2, \dots$. The environment consists of a single user and two sellers, denoted by A and B . In each period, the user selects a seller $y_t \in \{A, B\}$ and receives binary utility $u_t \in \{0, 1\}$.

Seller B is a transparent outside option. If the user selects Seller B, utility is Bernoulli with known success probability $p_0 \in (0, 1)$:

$$\mathbb{P}(u_t = 1 \mid y_t = B) = p_0. \quad (1)$$

Seller A has two products, $\mathcal{X} = \{1, 2\}$. Conditional on being selected, Seller A privately assigns product $x_t \in \mathcal{X}$. Product i generates a successful outcome with probability p_i and costs Seller A c_i :

$$\mathbb{P}(u_t = 1 \mid y_t = A, x_t = i) = p_i. \quad (2)$$

We assume

$$0 < p_1 < p_2 \leq 1, \quad 0 \leq c_1 < c_2, \quad (3)$$

so product 2 is more likely to satisfy the user but is also more costly. Whenever Seller A is chosen, it receives fixed revenue $R > 0$. Its period profit is

$$\pi_t^A = \begin{cases} R - c_{x_t}, & y_t = A, \\ 0, & y_t = B. \end{cases} \quad (4)$$

2. User Learning and Demand

The user does not observe Seller A's product choice. Product quality affects the distribution of realized utility, but the user sees only the selected seller and the binary outcome.

ASSUMPTION 1 (User's Misspecified Belief). *After selecting Seller A, the user observes u_t but not x_t . The user therefore treats Seller A as a single product with an unknown Bernoulli success probability $\theta \in [0, 1]$.*

Let S_t and F_t be the cumulative successes and failures generated by Seller A before period t :

$$S_t = \sum_{\tau=1}^{t-1} \mathbb{1}\{y_\tau = A, u_\tau = 1\}, \quad (5)$$

$$F_t = \sum_{\tau=1}^{t-1} \mathbb{1}\{y_\tau = A, u_\tau = 0\}. \quad (6)$$

Starting from the uniform prior $\theta \sim \text{Beta}(1, 1)$, the posterior is

$$\theta \mid S_t, F_t \sim \text{Beta}(1 + S_t, 1 + F_t). \quad (7)$$

The corresponding posterior mean is

$$m(S_t, F_t) = \frac{S_t + 1}{S_t + F_t + 2}. \quad (8)$$

The user follows Thompson sampling. At the beginning of period t , the user draws

$$\tilde{\theta}_t \sim \text{Beta}(1 + S_t, 1 + F_t) \quad (9)$$

and selects Seller A if and only if $\tilde{\theta}_t \geq p_0$. Seller A observes the user's choices and the realized outcomes, and can therefore track (S_t, F_t) . At state (S, F) , the probability that Seller A is selected is

$$\begin{aligned} D(S, F) &\triangleq \mathbb{P}(\tilde{\theta}_t \geq p_0 \mid S_t = S, F_t = F) \\ &= \int_{p_0}^1 \frac{\theta^S (1 - \theta)^F}{B(1 + S, 1 + F)} d\theta, \end{aligned} \quad (10)$$

where $B(\cdot, \cdot)$ is the Beta function. Demand determines how often Seller A earns revenue and receives a new reputation signal.

3. Discounted Seller Problem

Let $\gamma \in (0, 1)$ be Seller A's discount factor. Seller A chooses a policy that maps the user's posterior state to a hidden product and maximizes

$$\mathbb{E} \left[\sum_{t=1}^{\infty} \gamma^{t-1} \mathbb{1}\{y_t = A\} (R - c_{x_t}) \right]. \quad (11)$$

Let $V(S, F)$ be the optimal infinite-horizon value. If the user chooses Seller B, Seller A earns no current profit, the state does not change, and calendar time still advances. The stationary Bellman equation is

$$\begin{aligned} V(S, F) = & \gamma[1 - D(S, F)]V(S, F) \\ & + D(S, F) \max_{x \in \{1, 2\}} \{R - c_x + \gamma[p_x V(S+1, F) + (1 - p_x)V(S, F+1)]\}. \end{aligned} \quad (12)$$

This equation characterizes the discounted model. The computation below does not substitute a finite-horizon policy for the discounted policy. Instead, it approximates the solution to eq. (12) by retaining a long discounted tail and then checks whether the early-period policy is stable as the computational horizon increases.

3.1. Exact Long-Horizon Backward Induction

For a computational horizon T , let $V_t^T(S, F)$ denote the discounted value from periods $t, \dots, T$, with

$$V_{T+1}^T(S, F) = 0. \quad (13)$$

The exact recursion is

$$\begin{aligned} V_t^T(S, F) = & \gamma[1 - D(S, F)]V_{t+1}^T(S, F) \\ & + D(S, F) \max_{x \in \{1, 2\}} \{R - c_x + \gamma[p_x V_{t+1}^T(S+1, F) + (1 - p_x)V_{t+1}^T(S, F+1)]\}. \end{aligned} \quad (14)$$

At period t , every reachable state satisfies $S + F \leq t - 1$. Backward induction therefore covers the complete triangular state space for the stated horizon. Only adjacent value layers are stored in memory. In particular, the algorithm does not truncate at a fixed value of $S + F$, make a count boundary absorbing, or project simulated states back onto a smaller state space.

Conditional on the user choosing Seller A, define

$$Q_{t,x}^T(S, F) \triangleq R - c_x + \gamma[p_x V_{t+1}^T(S+1, F) + (1 - p_x)V_{t+1}^T(S, F+1)]. \quad (15)$$

Let the raw continuation difference be

$$G_t^T(S, F) \triangleq V_{t+1}^T(S+1, F) - V_{t+1}^T(S, F+1) \quad (16)$$

and define the discounted continuation value gap

$$M_t^T(S, F) \triangleq \gamma G_t^T(S, F). \quad (17)$$

Writing $\Delta p = p_2 - p_1$ and $\Delta c = c_2 - c_1$, the action-value difference is

$$Q_{t,2}^T(S, F) - Q_{t,1}^T(S, F) = -\Delta c + \Delta p M_t^T(S, F). \quad (18)$$

Consequently, product 2 is optimal if and only if

$$M_t^T(S, F) \geq \frac{\Delta c}{\Delta p}. \quad (19)$$

If the raw gap G_t^T is reported instead, its threshold is $\Delta c/(\gamma \Delta p)$. The distinction matters when interpreting value gap plots; the reported object M_t^T already contains the discount factor.

4. Discounted Fluid Approximation

The exact dynamic program retains the full distribution of the posterior count process. A deterministic fluid approximation replaces random increments by their conditional expected drifts. This provides an interpretable long-run benchmark and avoids enumerating a growing triangular state space. It does not alter the information structure: the user still observes outcomes but never observes Seller A's product.

4.1. State Dynamics and Objective

Let $\tau \geq 0$ denote continuous calendar time and let $s(\tau)$ and $f(\tau)$ be fluid success and failure counts. Extend demand to nonnegative real counts:

$$\mathcal{D}(s, f) \triangleq \int_{p_0}^1 \frac{\theta^s (1 - \theta)^f}{B(1 + s, 1 + f)} d\theta. \quad (20)$$

Let $a(\tau) \in [0, 1]$ be the relaxed intensity of product 2 and define

$$q(a) = p_1 + a \Delta p, \quad (21)$$

$$c(a) = c_1 + a \Delta c. \quad (22)$$

The expected count dynamics are

$$\dot{s}(\tau) = \mathcal{D}(s(\tau), f(\tau)) q(a(\tau)), \quad (23)$$

$$\dot{f}(\tau) = \mathcal{D}(s(\tau), f(\tau)) [1 - q(a(\tau))]. \quad (24)$$

Set $r = -\log \gamma$, so the continuous-time discount factor over τ periods is $e^{-r\tau} = \gamma^\tau$. For horizon H , the discounted fluid objective is

$$J_H^{\text{fl}}(a) = \int_0^H e^{-r\tau} \mathcal{D}(s(\tau), f(\tau)) [R - c(a(\tau))] d\tau. \quad (25)$$

This is a mean-drift closure, not an exact reformulation: in general, $\mathbb{E}[D(S_t, F_t)] \neq D(\mathbb{E}[S_t], \mathbb{E}[F_t])$.

4.2. Fluid Optimality Condition

Let $\mathcal{V}(\tau, s, f)$ be the optimal fluid value from τ to H , discounted relative to time τ , with $\mathcal{V}(H, s, f) = 0$. Its Hamilton–Jacobi–Bellman equation is

$$-\partial_\tau \mathcal{V} + r\mathcal{V} = \mathcal{D}(s, f) \max_{a \in [0,1]} \{R - c(a) + q(a)\partial_s \mathcal{V} + [1 - q(a)]\partial_f \mathcal{V}\}. \quad (26)$$

The coefficient multiplying a inside the Hamiltonian is

$$-\Delta c + \Delta p(\partial_s \mathcal{V} - \partial_f \mathcal{V}). \quad (27)$$

Thus the relaxed solution is bang-bang except at indifference. Product 2 is optimal whenever

$$\partial_s \mathcal{V} - \partial_f \mathcal{V} \geq \frac{\Delta c}{\Delta p}. \quad (28)$$

The fluid model therefore preserves the discrete model’s central tradeoff: the costly product is used when the shadow value of replacing a failure by a success is sufficiently large. Discounting changes the value function and its derivatives through $r\mathcal{V}$, even though it does not appear separately in the threshold once the discounted shadow value has been computed.

4.3. Effective Observations and Long-Run Quality Mix

Define effective observations and the fluid posterior mean by

$$n(\tau) = s(\tau) + f(\tau), \quad m(\tau) = \frac{s(\tau) + 1}{n(\tau) + 2}. \quad (29)$$

Let $\widehat{\mathcal{D}}(n, m)$ denote eq. (20) written in this transformed state. Then

$$\dot{n}(\tau) = \widehat{\mathcal{D}}(n(\tau), m(\tau)), \quad (30)$$

$$\dot{m}(\tau) = \frac{\widehat{\mathcal{D}}(n(\tau), m(\tau))}{n(\tau) + 2} [q(a(\tau)) - m(\tau)]. \quad (31)$$

Demand determines the rate at which the seller collects observations, while the delivered quality determines the direction of belief updating. The factor $1/(n+2)$ captures belief rigidity as evidence accumulates.

If a constant control a is used and demand remains positive, dividing eq. (31) by eq. (30) gives

$$\frac{dm}{dn} = \frac{q(a) - m}{n + 2}. \quad (32)$$

Starting from the uniform prior, the posterior mean converges to $q(a)$. When $p_1 < p_0 \leq p_2$, the product-2 intensity that places limiting quality exactly at the outside option is

$$a_{\text{bdry}}(p_0) = \frac{p_0 - p_1}{p_2 - p_1}. \quad (33)$$

For $p_0 \leq p_1$, product 1 alone clears this benchmark. For $p_0 > p_2$, no feasible mix can sustain a limiting quality at least as high as the outside option. The boundary mix is a sustainability benchmark, not the solution to the discounted control problem: optimal investment also reflects current costs, discounting, posterior uncertainty, and the option value of future demand.

5. Numerical Study

5.1. Calibration and Computation

The numerical study uses the calibration in table 1. Product 2 raises the success probability by $\Delta p = 0.45$ and costs an additional $\Delta c = 0.60$. Hence the discounted continuation-gap threshold in eq. (19) is

$$\frac{\Delta c}{\Delta p} = \frac{4}{3}. \quad (34)$$

Table 1 Numerical calibration.

Parameter	p_1	p_2	c_1	c_2	R
Value	0.35	0.80	0.05	0.65	1.00

The main patient-seller experiment sets $\gamma = 0.999$, solves eq. (14) through $T = 2000$, and simulates the first 250 periods for 400 independent replications at each $p_0 \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$. The final-window statistics use periods 201–250.

For $\gamma = 0.98$, the default horizon $T = 700$ satisfies $\gamma^T = 7.22 \times 10^{-7}$, so the omitted discounted tail is negligible under a 10^{-6} criterion. For $\gamma = 0.999$, however, $\gamma^{2000} = 0.1352$; a literal $\gamma^T < 10^{-6}$ requirement would need $T = 13809$. Thus the $T = 2000$ exercise should be interpreted as a large-horizon policy experiment, not as a strict value-tail approximation. The appropriate diagnostic is stability of the early policy under further horizon increases.

5.2. Simulation Results

Figure 1 reports market share, product use, posterior beliefs, and profit over the simulation window. The late-window results are summarized in table 2.

Exact discounted simulated paths

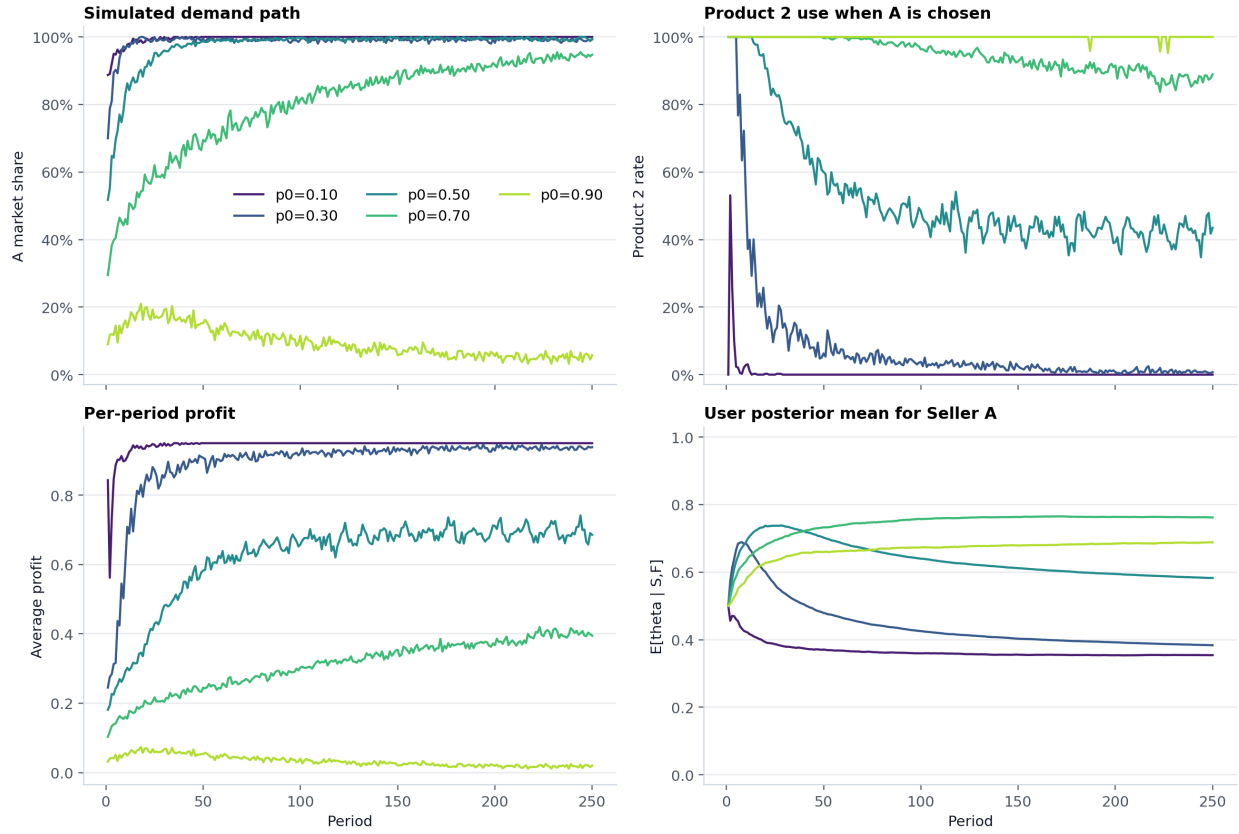


Figure 1 Simulation outcomes under the exact discounted policy with $\gamma = 0.999$ and $T = 2000$. Lines aggregate 400 independent replications for each outside-option quality.

Table 2 Late-window simulation results, periods 201–250.

p_0	Fluid boundary a_{bdry}	Seller A share	Product-2 rate	Posterior mean
0.10	0	1.000	0.000	0.354
0.30	0	0.992	0.009	0.387
0.50	0.333	0.994	0.420	0.588
0.70	0.778	0.935	0.883	0.763
0.90	infeasible	0.051	0.998	0.686

The product-2 rate is conditional on Seller A being selected. The fluid boundary is the minimum quality-mix benchmark in eq. (33); it is not an optimality prediction.

When $p_0 \in \{0.1, 0.3\}$, the low-cost product already has quality at least as high as the outside option, and reputation investment vanishes in the late window. At $p_0 = 0.5$, Seller A is retained in 99.4% of interactions and uses product 2 in 42.0% of its selected periods. At $p_0 = 0.7$, stronger quality investment keeps Seller A's market share near 93.5%. The optimal product-2 rates exceed the fluid boundary at both interior cutoffs, consistent with the fact that equality only centers the

limiting posterior at the user's cutoff. At $p_0 = 0.9 > p_2$, even product 2 cannot match the outside option. Seller A uses it almost whenever selected, but its late market share falls to 5.1%.

Figure 2 compares the simulated product-2 rates with the fluid quality-maintenance benchmark. The comparison is qualitative: stochastic posterior paths, endogenous selection, and discounted option value are absent from the deterministic boundary calculation.

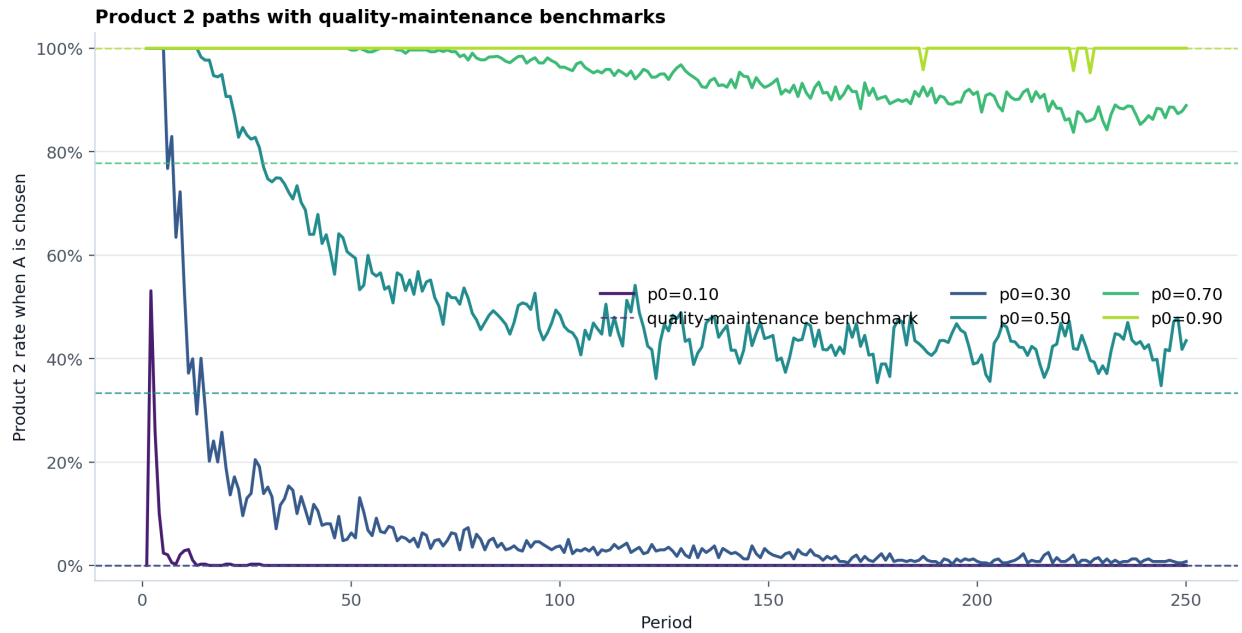


Figure 2 Product-2 use under the exact discounted policy and the clipped fluid quality-mix benchmark. The benchmark indicates the product mix required to match the outside-option quality when such a mix is feasible.

Finally, fig. 3 displays early-period policy slices. Yellow states use product 1 and teal states use product 2. The switching region is not a rule based only on the posterior mean: the total number of observations and the remaining discounted continuation value also affect whether an additional success is worth its cost.

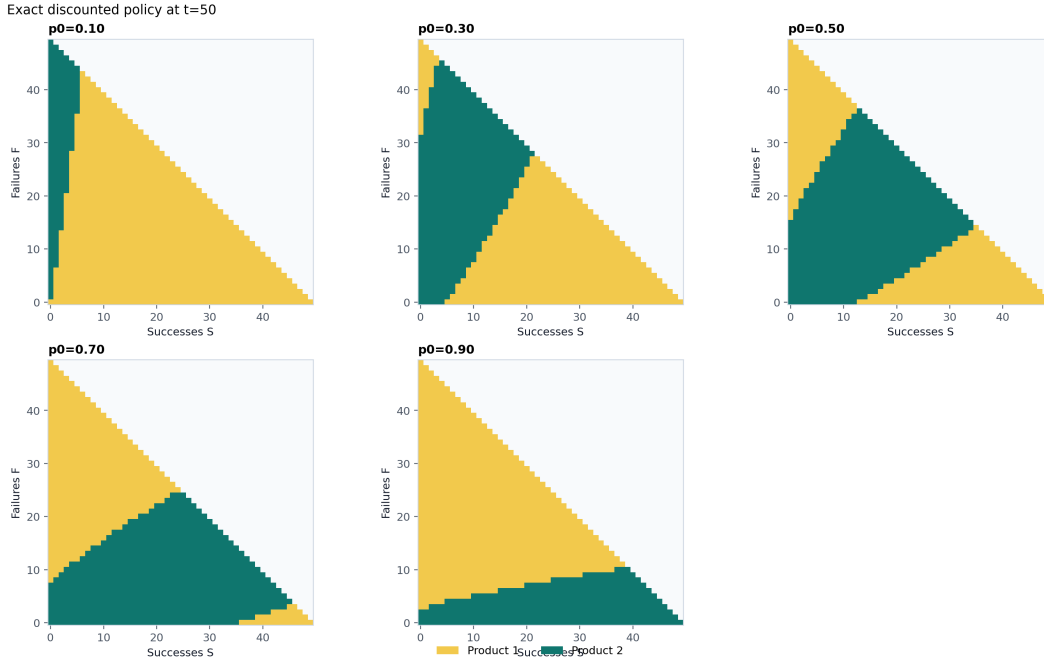


Figure 3 Exact discounted product policy over reachable posterior states. The policy uses product 2 precisely where the discounted continuation value gap exceeds $4/3$.

6. Discussion

The model separates three objects that are easy to conflate. Demand $D(S, F)$ determines how frequently Seller A earns revenue and updates the user’s belief. The posterior state determines how responsive reputation is to another outcome. The discounted gap $M_t^T(S, F)$ measures the economic value of turning the next failure state into the next success state. Product 2 is chosen only when that last object is large enough to cover its incremental cost.

The fluid approximation preserves this mechanism through a shadow-value condition and adds a transparent benchmark for the long-run product mix. However, it suppresses path dispersion and is least reliable near $m = p_0$, where small posterior fluctuations sharply change Thompson-sampling demand. It should therefore complement, rather than replace, exact backward induction and stochastic simulation.

7. Conclusion

Reputation investment varies nonmonotonically with the user’s outside option. When the outside option is weak, the cheap product is sufficient. At intermediate cutoffs, the seller uses the costly product to keep its posterior reputation above the user’s threshold. When the outside option dominates even the high-quality product, quality investment cannot prevent demand loss. The exact discounted recursion and the fluid benchmark expose these forces from complementary perspectives.